

GBIF Biodiversity Data Gaps

Integrated Assessment: Spatial, Temporal, and Taxonomic Dimensions

gbifgaps

February 11, 2026

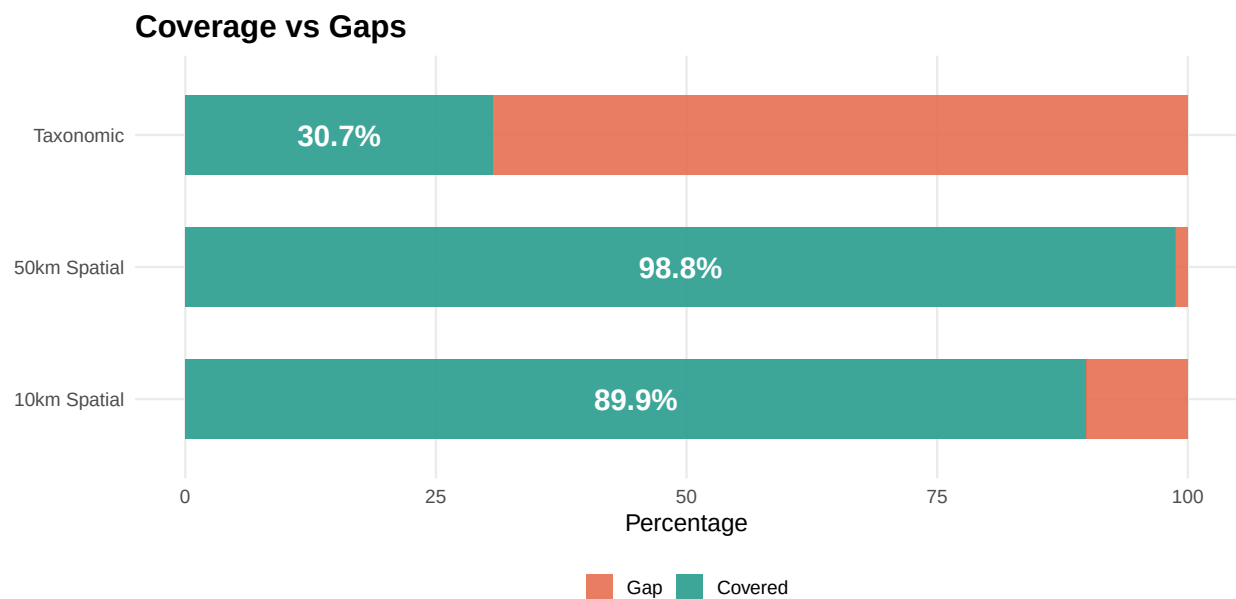
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Executive Summary

This report presents a comprehensive assessment of biodiversity data gaps in the national GBIF occurrence database, integrating spatial, temporal, and taxonomic dimensions to inform strategic priorities for data collection and mobilization.

Dimension	Current Status	Assessment
Spatial Coverage (10km)	6,284 / 6,988 cells (89.9%)	Good
Spatial Coverage (50km)	332 / 336 cells (98.8%)	Excellent
Temporal Range	N/A years (1646-2025)	Good
Data Recency	5 months median; 23.6% cells >5yr stale	Recent
Taxonomic Coverage	51,525 / 168,043 taxa (30.7%)	Low
Priority Taxa	1,351 taxa requiring attention	Action Required



1. Spatial Gaps

1.1 Geographic Coverage

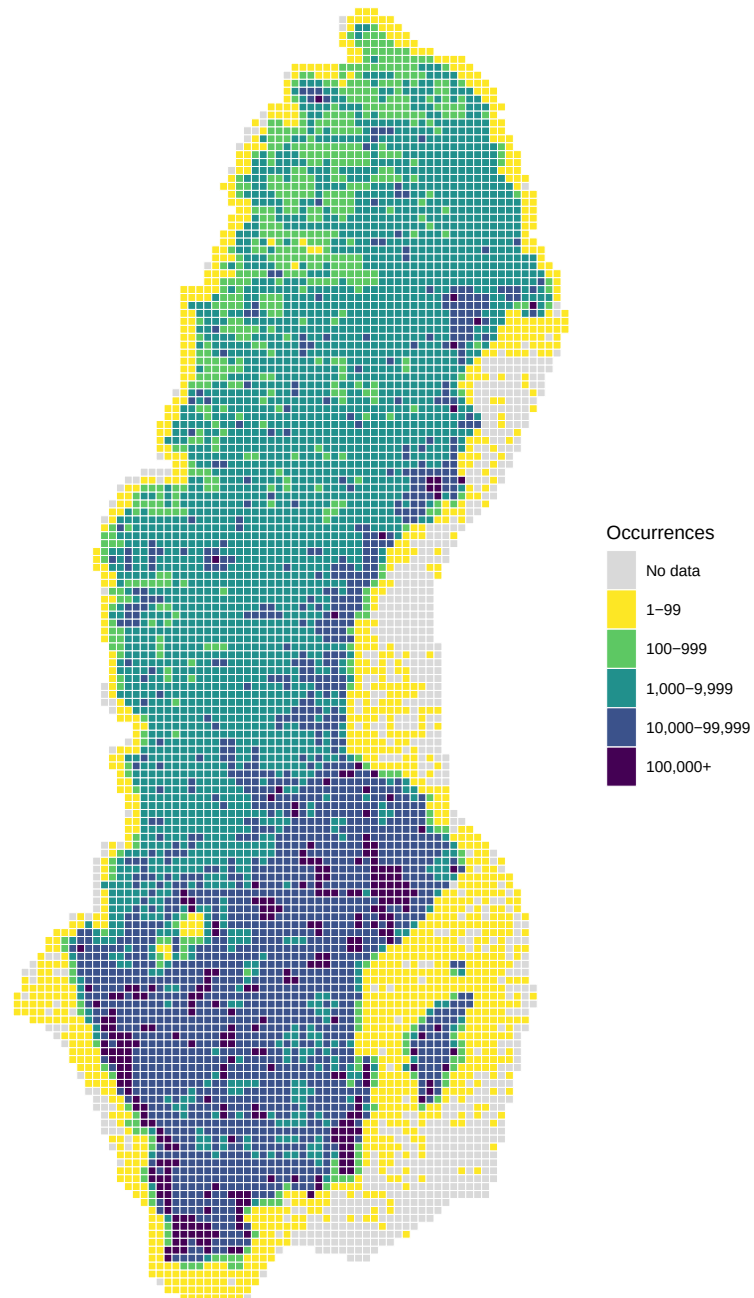
Required columns not found in comparison_grid_resolutions.csv

Available columns: grid, n_cells_total, n_cells_with_data, n_cells_zero, pct_coverage, total_occurrence

1.2 Geographic Distribution

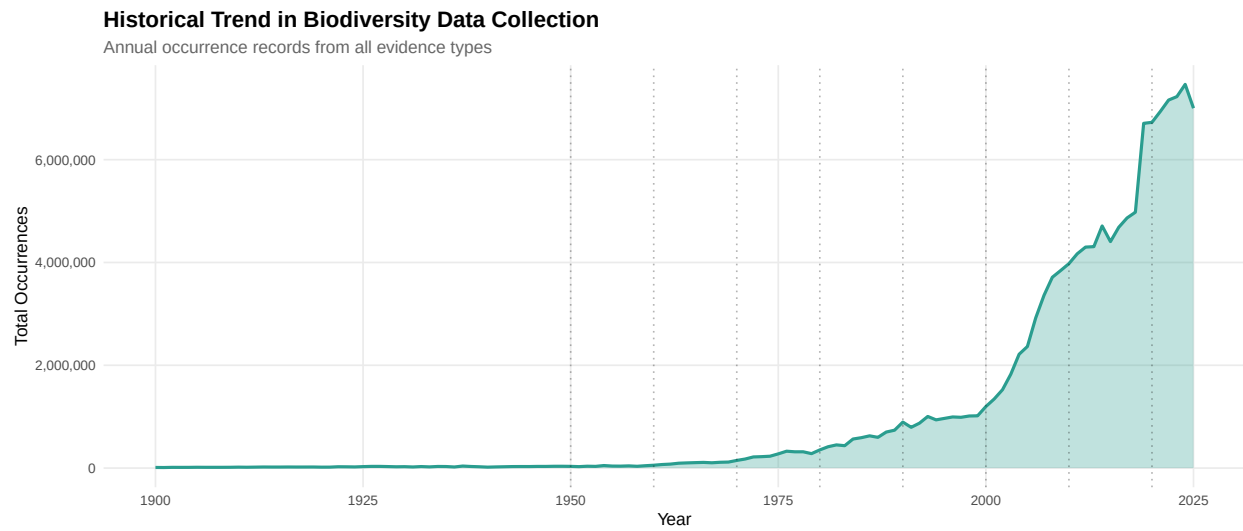
Geographic Distribution of Occurrence Data

10km EEA Grid



2. Temporal Gaps

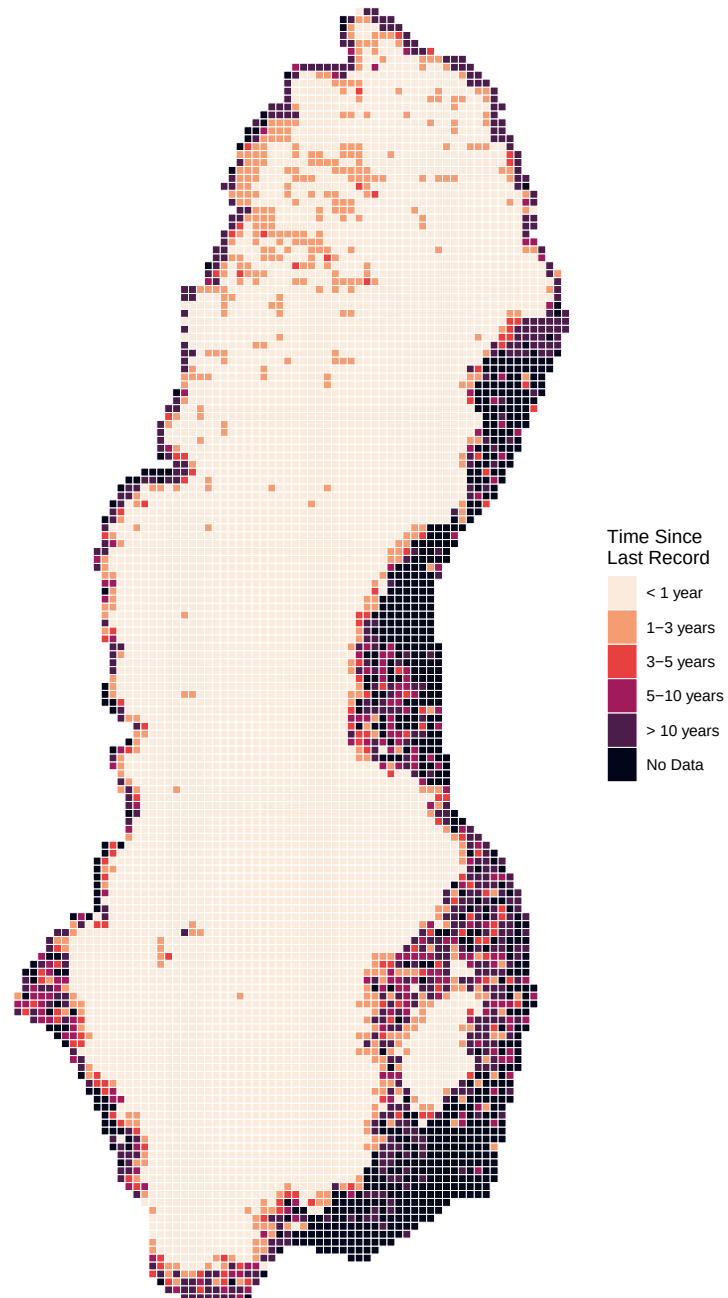
2.1 Historical Trends



2.2 Data Recency

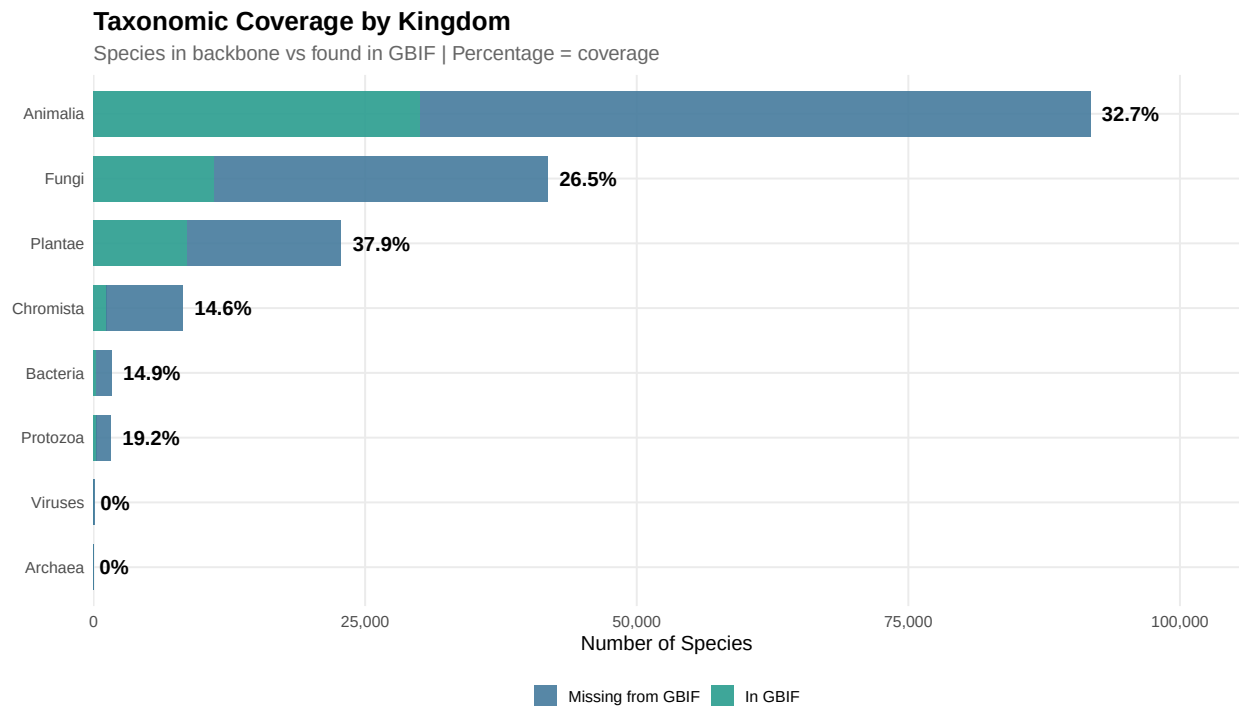
Data Staleness Across Study Area

Time since last occurrence record per 10km cell



3. Taxonomic Gaps

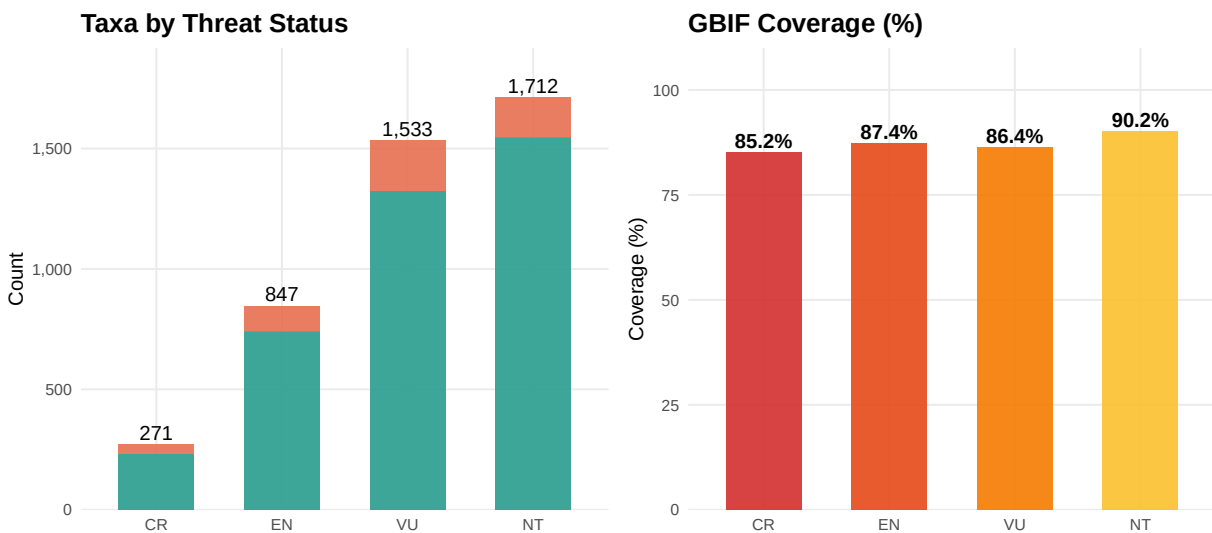
3.1 Coverage by Kingdom



3.2 Coverage by Threat Status

Threatened Species Coverage in GBIF

CR = Critically Endangered, EN = Endangered, VU = Vulnerable, NT = Near Threatened, LC = Least Concern



4. Priority Recommendations

4.1 Key Findings

Summary of Data Gaps

Spatial Coverage: 89.9% of the country's 10km grid cells have occurrence data.

Data Recency: 23.6% of cells have not been surveyed in over 5 years.

Taxonomic Coverage: 30.7% of backbone species have GBIF records.

Missing Threatened Species: 147 Critically Endangered and Endangered species lack GBIF records.

4.2 Priority Actions

Priority	Category	Action	Target
High	Spatial	Survey zero-coverage grid cells	0 cells
High	Taxonomic	Monitor CR and EN species lacking GBIF records	147 species
Medium	Temporal	Re-survey cells with >5 years since last observation	1,805 cells
Medium	Taxonomic	Digitize museum specimens for underrepresented orders	Focus on low-coverage taxonomic groups
Low	Spatial	Expand citizen science programs to low-coverage regions	Remote and mountainous regions

5. Data Sources & Methods

GBIF Occurrence Cubes

- 10km grid: <https://doi.org/10.15468/dl.wzv3uc>
- 50km grid: <https://doi.org/10.15468/dl.qyp3uw>

Taxonomic Reference

- National Taxonomy Backbone: <https://doi.org/10.15468/j43wfc>
- National Red List: <https://doi.org/10.15468/jhwkpw>

Analysis Date: 2026-02-10

For questions or more information, please contact your GBIF node.